

Parallelograms — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

In a parallelogram, opposite angles are:

- (a) supplementary (b) complementary (c) equal (d) unequal

Q2 • MCQ**[1]**

If the diagonals of a quadrilateral bisect each other, it must be a:

- (a) trapezium (b) parallelogram (c) kite (d) none of these

Q3 • FILL IN THE BLANK**[1]**

In a parallelogram, one angle is 65° . Its adjacent angle is _____ degrees.

Q4 • MCQ**[1]**

Which is **NOT** always true for a parallelogram?

- (a) Opposite sides equal (b) Diagonals equal (c) Opposite angles equal
(d) Diagonals bisect each other

Q5 • TRUE / FALSE**[1]**

In a parallelogram, the diagonals bisect each other.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): If the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Reason (R): In a parallelogram, diagonals always bisect each other.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

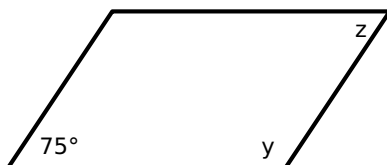
[2]

In parallelogram ABCD, $\angle A : \angle B = 2 : 3$. Find $\angle A$ and $\angle B$.

Q8 · FIGURE

[2]

Find $\angle y$ and $\angle z$ in the parallelogram below.



Q9

[2]

The diagonals of parallelogram PQRS meet at O. If $PO = 4$ cm and $QO = 3$ cm, find PR and QS. Give reasons.

Long answer (3 marks each)

Q10

[3]

One angle of a parallelogram is three times its adjacent angle. Find all four angles and state the properties you used.

Q11

[3]

In parallelogram ABCD, $\angle A = 80^\circ$. Find $\angle B$, $\angle C$ and $\angle D$, giving a reason for each.

Q12

[3]

A convex quadrilateral has interior angles 90° , 75° and 85° . Find the fourth angle and explain why this quadrilateral cannot be a parallelogram.

Total: 21 marks · Check your work against the [answer key](#).